AI Forward Deployed Engineer

The AI Architect & FDE Course



Week 3 Day 4

3 weeks to Forward Deployed Engineer

Theory Practical Projects

Week 3: Become an AI FDE

- AI Solutions Architect
- Security, Guardrails, API Gateways
- Enterprise Context Layer & Integrations
- **Production Deployment**
- Capstone

Week 2: Become an AI Architect

AI ARCHITECT

- Architecture, Evals, Observability
- Agentic RAG, MCP, Memory
- Durable State & Human in the Loop
- Agent Teams, Swarms, Fleet
- Self-Improvement, RSI

Week 1: Become an AI Engineer

AI ENGINEER

- Path to FDE
- Frontier LLMs & Agents
- Coding Agents & ADLC
- Open-Source, Local, Token Cost
- Harness Engineering

Taking stock

You can already:

- ✓ Deliver solutions that measurably solve business problems with evals and observability
- ✓ Create durable, scalable agents with memory and a human in the loop
- ✓ Drive an AI implementation from discovery to delivery and maintenance
- ✓ Create an enterprise agent architecture with a gateway, context layer & memory

By the end of today, you'll be able to:

- ○ Deploy an agent to a managed runtime in the cloud
- ○ Prepare an agent for production with settings, secrets and limits
- ○ Differentiate between a managed agent runtime and containers
- ○ Architect an agent solution with durable execution and human in the loop

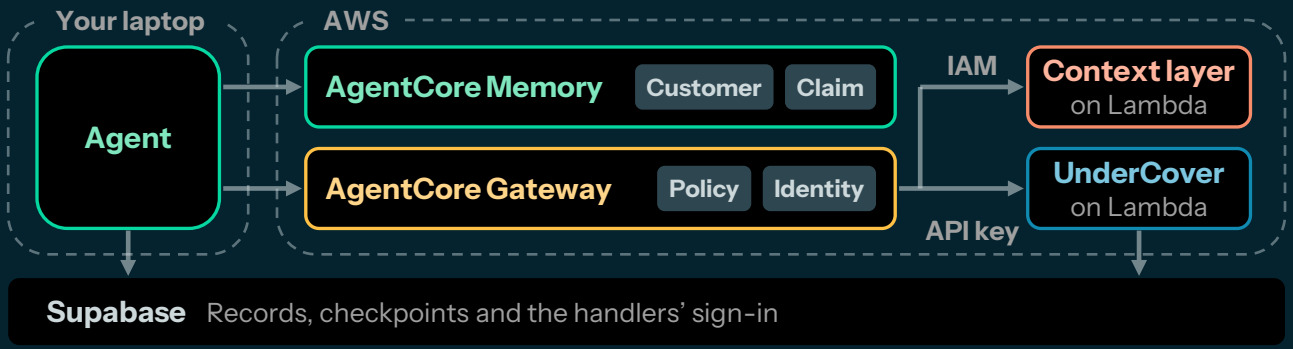
Four sources of data and integration

Each one has a different owner

	Who writes it	The agent
 **Gateway** Rules for tool calls	**The security team** Policies like the payout limit	**Cannot change it** Every call goes through it
 **Context layer** Data from other systems	**The business teams** Data governance model	**Reads it** As tools through the gateway
 **Memory** The agent's own notes	**The agent** Separate for each customer	**Reads and writes it** With its memory tools
 **Database** The system of record	**The app and the agent** Records and checkpoints	**Reads and writes it** Through the app's API

Yesterday

We ran UnderCover with its context layer on AWS



Configuring for production

The housekeeping associated with deployment

Settings from the environment

In UnderCover

The app, Gateway and Memory addresses are set on the runtime

Secrets in a vault

In UnderCover

The keys and the database password are in AgentCore Identity

Who may call it

In UnderCover

Only the app's IAM role may invoke the agent

Limits

In UnderCover

A session ends after 60 idle seconds

Logs and cost

In UnderCover

CloudWatch for logs and `monitor.py` for spend

Where the agent runs

3 ways to host an agent in production

	Examples	What you get
Your own infrastructure	**AWS Lambda** ECS, Cloud Run or Kubernetes Render	Any framework and model State and scaling are yours
A managed runtime	**AWS AgentCore Runtime** Google Agent Runtime Microsoft Foundry LangSmith Deployment	Any framework and model Memory and identity built in
The model lab runs the agent	Claude Managed Agents OpenAI Agents API	The lab's own harness Only the lab's models

The AgentCore CLI

The commands that take an agent from your laptop to AWS

1. **Install it once** Pinned to the lab's version
   ```
   npm install -g @aws/agentcore@1.0.0-rc.5
   ```
2. **Start a project** A LangGraph agent and its config
   ```
   agentcore create --template agent-python-langchain
   ```
3. **Run it on your laptop** A local server to try it
   ```
   agentcore dev
   ```
4. **Deploy it to AWS** A zip build on AgentCore Runtime
   ```
   agentcore deploy
   ```
5. **Call it** Or from Python with boto3
   ```
   agentcore invoke --runtime agent
   ```
6. **Tear it down** Then deploy again to delete it from AWS
   ```
   agentcore remove all
   ```

Lab 1: AgentCore Runtime

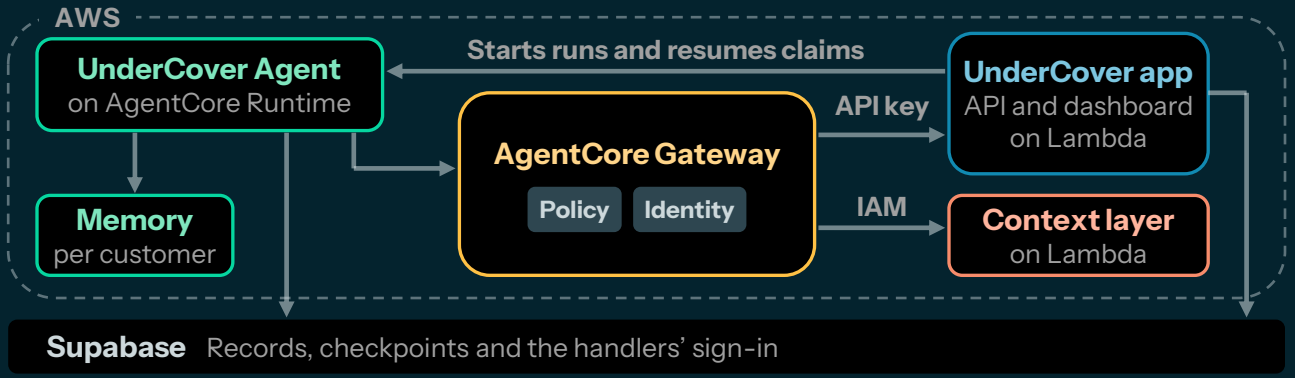
A small agent deployed, paused and resumed



1. **Create and deploy a hello world**
   With `agentcore create`
2. **Run the LangGraph agent locally**
   With `agentcore dev`
3. **Deploy it to Runtime**
   Zip and deploy
4. **Call it twice**
   A cold start, then a warm one
5. **Pause and resume**
   A new session picks up from Supabase
6. **Tear down**
   The account setup stays for Lab 2

UnderCover by the end of today

The agent joins the app on AWS



Agent on Runtime
One session for each dispatch or evaluation

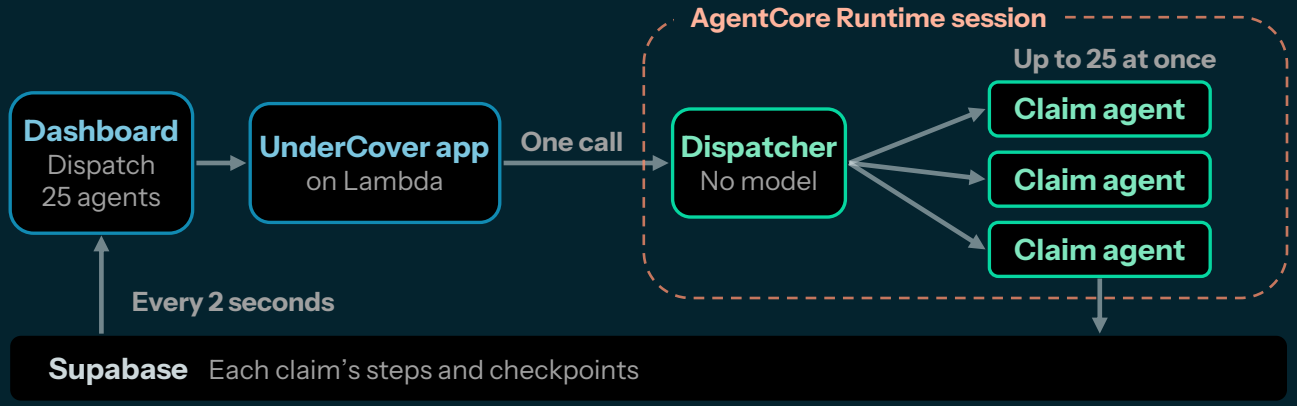
Claim approval
Sign off, or send it back with a note

UnderCover app
Starts runs and resumes claims

Supabase
Keeps the checkpoints between sessions

The agent on Runtime

One session works a whole dispatch



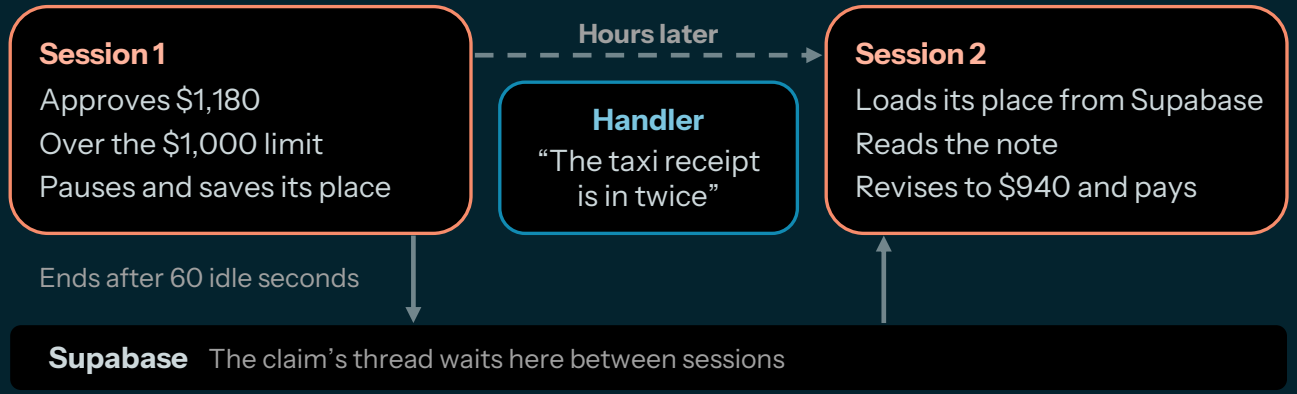
Replies at once
The app's call returns as soon as the work starts

Keeps working
The session runs until the work is done

One process
Today's code, with one pool of connections

Sending a claim back

A handler's note resumes the claim in a new session

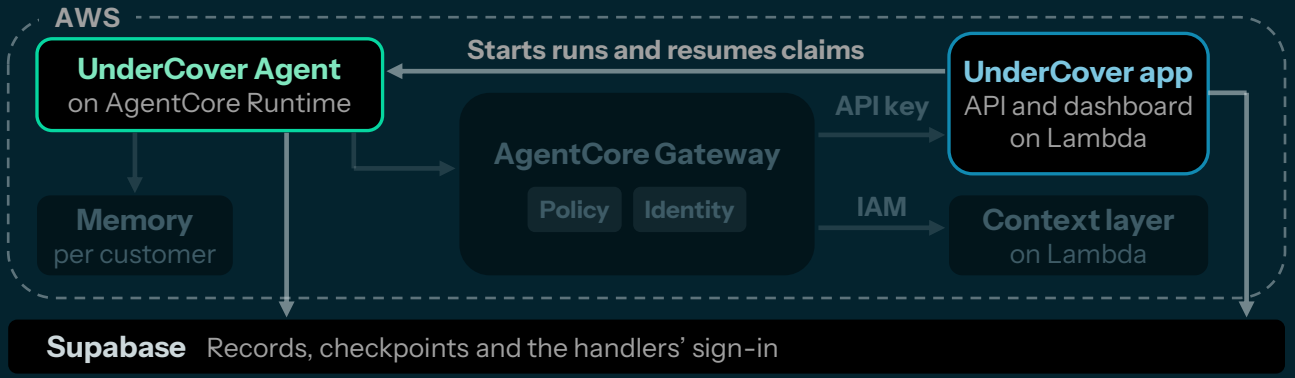


Sign off The app pays, as it does today

Send back The agent gets the note and decides again

Upgrading the product

Four changes to the app and the agent



1. **The claim approval comes back**
   A payout over the limit waits for a handler
2. **Sign off or send it back**
   A handler's note goes back to the agent
3. **The agent as an AgentCore app**
   The same app on a laptop and on Runtime
4. **The app keeps one run at a time**
   Two sessions can run at once on Runtime

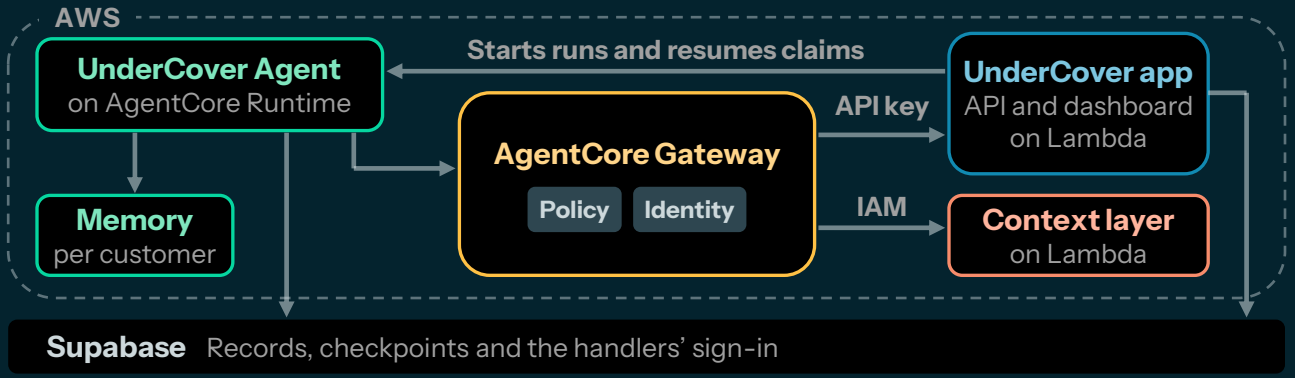
Building it with Claude Code

The change is written up in labs/week3day4_product_RUNTIME.md

1. **Copy the brief into docs**
   `cp labs/week3day4_product_RUNTIME.md docs/RUNTIME.md`
2. **Ask Claude Code for a plan**
   Please plan the Runtime changes in docs/RUNTIME.md
3. **Let it build and commit each phase**
   Please implement the plan for the Runtime changes
4. **Or check out my version**
   `git checkout week3day4-start` or `git checkout week3day4-end`

Lab 2: UnderCover on Runtime

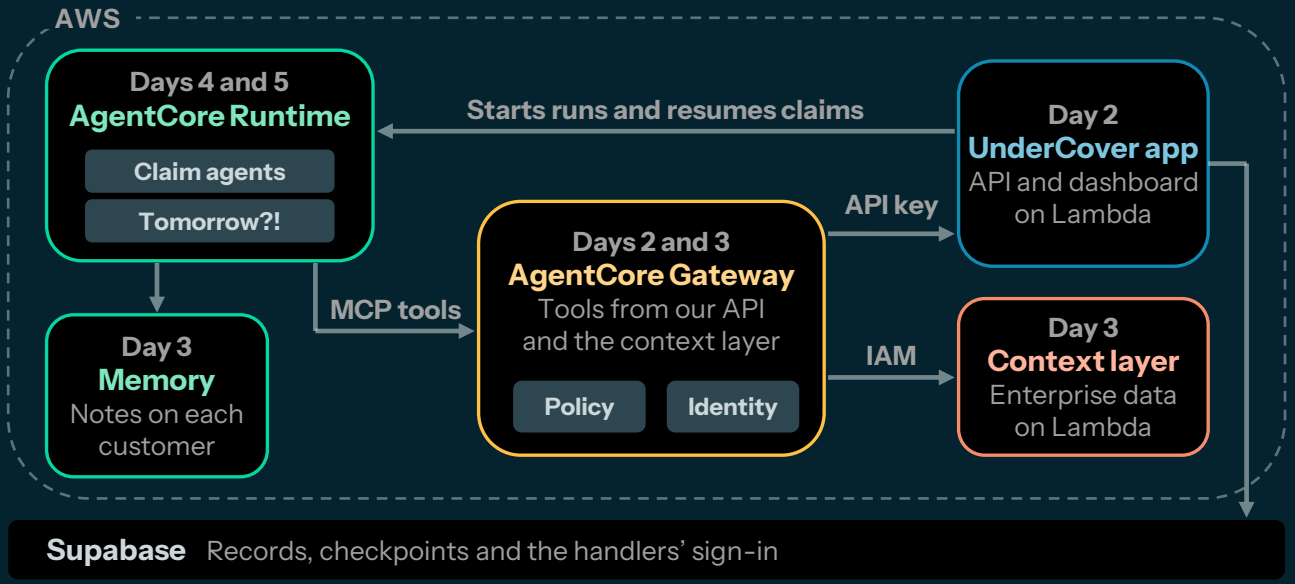
Deploying the agent beside the app



1. **Deploy the new app**
   The same function with the new routes
2. **Bring back Day 3's pieces**
   The Gateway, the Policy and the Memory
3. **Deploy the agent**
   `agentcore deploy` with its settings
4. **Point the app at the agent**
   Its role may now invoke the runtime
5. **Run a swarm and send a claim back**
   25 agents, then a handler's note
6. **Tear down**
   Delete the agent, Gateway and Memory

UnderCover is deployed!

But is there more?



Models still come from OpenRouter

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93% Complete